MILITARY SPECIFICATION

TOWER AB-305( ) FPS

1. SCOPE.

1.1 Scope. This specification covers one type of extension tower designated as Tower AB-305( ) FPS, herein after referred to as the extension tower.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

FF-H-106	Hardware, Builders, Locks and Door Trim
FF-H-116	Hinges, Hardware, Builders
FF-B-575	Bolts, Square and Hexagon
QQ-S-775	Sheet Steel, Carbon, Zinc Coated
QQ-S-781	Strapping, Steel and Seals
SS-C-153	Cement, Bituminous, Plastic
PPP-B-601	Box, Wood, Cleated, Plywood
PPP-B-621	Box, Wood, Nailed and Corner Locked

Military

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-S-3950	Switch, Toggle
MIL-E-4158	Electronic Equipment, Ground, General Requirements for
MIL-C-6136	Conduit, Electrical Flexible Shielded Aluminum Alloy for Aircraft Installations
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-C-18969	Caulking Compounds, Metal and Wood Seam
MIL-C-38427	Coating Kit, Top Coat, Pigmented, for Application Over Inorganic Zinc Rich Primer Coated Steel Surfaces

FSC 5445

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipping and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Procedure for Selection of
MIL-STD-882	Systems Safety Program, for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods

DRAWINGS

Air Force

D/L 57A54070 and Details

Steelcraft Manufacturing Co.
D/L 33780 and Details

(Copies of applicable documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A-36	Specification for Structural Steel
A-123	Specification for Zinc-Coating (Hot Dip) on Products Fabricated from Rolled, Pressed or Forged Steel Shapes, Plates, Bars and Strips
A-153	Specification for Zinc-Coating (Hot Dip) on Iron and Steel Hardware
A-325	Specification for High Strength Carbon Steel Bolts for Structural Joints, Including Suitable Nuts and Plain Washers
A-361	Specification for Zinc-Coated (Galvanized) Iron or Steel Roofing Sheets

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A-446	Specification for Zinc-Coated (Galvanized) Steel Sheets of Structural Quality, Coils and Cut Lengths (Grade A).
A-449	Specification for Quenched and Tempered Steel Bolts and Studs
A-490	Specification for Quenched and Tempered Alloy Steel Bolts for Structural Steel Joints
A-525	Specification, General Requirements for Delivery of Zinc Coated Iron and Steel Sheets, Coils and Cut Lengths Coated by the Hot Dip Method

(Copies of ASTM specifications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY HANDBOOK

AWS	Specification Covering Structural Welding and Qualification of Welders

(Copies of AWS Handbook may be obtained from American Welding Society, United Engineering Center, 345 East 47th Street, NY, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Copies of Uniform Freight Classification Rules may be obtained from the Committee at 202 Union Station, 516 West Jackson Blvd, Chicago, IL 60606.)

AMERICAN TRUCKING ASSOCIATION (ATA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc. (ATA), Traffic Dept., 1616 P Street, NW, Washington, DC 20402.)

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Occupational Safety and Health Act (OSHA) Standards with Supplemental Revisions

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

NATIONAL ELECTRICAL CODE

(Applications for copies should be addressed to Underwriters' Laboratories, Inc., 207 E Ohio St., Chicago, IL 60611.)

3. REQUIREMENTS

3.1 First article testing. This specification makes provision for first article testing.

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3.2 Design and fabrication. Design and fabrication of the tower shall be in accordance with Air Force D/L 57A54070 and details with the addition of the following:

3.2.1 Stairway. The contractor shall fabricate and furnish a stairway in accordance with Steelcraft Manufacturing Company D/L 33780 and details.

3.2.2 Hatch door. The contractor shall furnish a hatch door to fit over the entrance opening for the basic tower. The door shall support its own weight. The door shall be light weight, consistent with materials specified herein and shall be provided with a handle for opening from above.

3.2.2.1 Hatch railing. A railing with gate approximately three feet high shall be provided around the hatch.

3.2.3 Selection of standards and specifications. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 General. Materials shall be as specified herein. Materials not definitely specified shall be selected by the supplier and shall be subjected to all the requirements of this specification.

3.3.2 Steel.

3.3.2.1 Structural steel. Steel for structure and framing members having a thickness greater than 0.060 inch, shall conform to ASTM Standard A-36.

3.3.2.2 Steel siding and roofing sheets. All siding and roofing sheets shall be in accordance with ASTM designation A-361 or A-446 Grade A. Zinc coated sheet steel in accordance with QQ-S-775 is acceptable.

3.3.3 Fasteners. Bolts for joining main structural frame members shall be high strength bolts conforming to ASTM designation A-325, A-449 or A-490 and shall be furnished with suitable nuts and plain hardened washers. Other bolts shall be in accordance with FF-B-575.

3.3.4 Sealant. Sealant for making watertight the joints of wall and roof panels shall be in accordance with MIL-C-18969. Optional sealant shall be bituminous plastic cement in accordance with SS-C-153.

3.3.5 Extra material. The supplier shall furnish an additional 5 percent minimum of bolts, nuts, washers and other fasteners, electrical fittings, sealant material and one (1) gallon of paint. This additional material shall be of the same type and character supplied for the assembly of the tower in the field.

3.3.6 Electrical. Electrical wiring and lighting shall be in accordance with referenced drawings and as specified herein. The design, material and installation shall meet or exceed the minimum requirements of the Underwriters' Laboratories, National Electrical Code.

3.3.6.1 Switches.

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3.3.6.1.1 Toggle. Toggle switches shall be in accordance with MIL-S-3950.

3.3.6.1.2 Interlock switches. Interlock switches shall be of the two piece type wherein the electrical circuit is broken by the separation of the two parts. Conversely the circuit shall be established by the mating of the two parts.

3.3.6.2 Conduit. Flexible shielded conduit of the amount and size shown on the drawings shall be in accordance with MIL-C-6136.

3.3.6.3 Plugs and receptacles. Connectors for alternating or direct power input connections shall be in accordance with MIL-E-4158.

3.3.7 Doors. Doors of the number and location shown on the referenced drawings shall be furnished with the tower. Doors shall be fabricated of steel, industrial type not less than 20 gage sheeting as specified herein (see 3.3.2.2). The door shall be insulated and sound deadened with a suitable material laminated to the inside with high strength adhesive. Each door shall be furnished complete with steel frame, threshold, weather stripping and full morticed prime painted hinges in accordance with FF-H-116. Panic hardware and all other hardware necessary for the installation of the complete door shall be furnished and shall be in accordance with FF-H-106.

3.3.8 Windows. Windows shall be of the number and location shown on the drawing and shall be of the type which open one-half the window area and lock in place. Windows shall be furnished with insect screens and shall have green tinted glass panes.

3.3.9 Ladder. A steel ladder shall be furnished and shall meet the requirements of OSHA Standards, Vol I, Para 1910.27, Fixed Ladders.

3.3.9.1 Safety rails. Safety rails to be mounted on the ladder shall be supplied with the tower. These safety rails shall have provisions to receive a safety belt clevis which immediately locks upon taking the suspended weight of a man. The safety rail shall meet the requirements of OSHA Standard, Para 1910.27(d)(5), Ladder Safety Devices. All necessary hardware shall be provided with this safety rail so that it may be mounted to the rungs of the ladder in accordance with the manufacturer's instructions. At least two leather belts, with the required sleeves and clevises for two complete sets of safety belts shall be furnished with each safety rail. The ladder safety system, including safety rail and belts, shall be designed to support a 200 pound man with a safety factor of 3.

3.4 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment.

3.5 Welding. All welders shall be qualified in accordance with standard qualification procedures as specified in the American Welding Society Handbook. All welding shall be done in accordance with accepted practices of the American Welding Society. There shall be no welding required for field erection.

3.6 Marking. Identification marking shall be in accordance with MIL-STD-130.

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3.6.1 Identification of members. All structural members shall be marked so as to permit ready identification by reference to erection and assembly drawings. Marking shall be accomplished by depth stamping, shall be completed prior to finishing of the steel and shall be legible after finishing. Depth stamping shall in no way flatten or deform the round cross section of tubular members.

3.7 Finishing.

3.7.1 Galvanizing. All structural steel shall be galvanized in accordance with ASTM Standard A-123. Hardware such as bolts, nuts, washers, etc., shall be galvanized in accordance with ASTM Standard A-153 and sheet metal not otherwise zinc-coated shall be galvanized in accordance with ASTM Standard A-525.

3.7.2 Painting. The tower shall be finished in accordance with MIL-F-14072. The finish color shall be lusterless gray number 35237 or white number 37875 in accordance with FED-STD-595. Finish paint shall conform to MIL-C-38427, Type III, Class 1.

3.8 Dissimilar metals. Unless suitably protected against electrolyte corrosion, dissimilar metals as defined by MIL-STD-889 shall not be used in initimate contact with each other.

3.9 Safety.

3.9.1 Category III and IV hazards. All equipment furnished under this specification shall be analyzed for category III and IV hazards as defined by MIL-STD-882. Any hazard so identified shall be brought to the attention of the procuring agency along with recommended action for satisfying safety requirements. Action order of precedence shall be as specified in MIL-STD-882, para 5.6.

3.10 Workmanship. The tower shall be neatly finished and all burrs or rough edges shall be removed. Welds shall be sound, smooth and essentially free from pits fractures burn through or fissures. Bolt and rivet holes shall be accurately punched or drilled and shall have burrs removed. Lockwashers shall be provided in accordance with standard commercial practice, and all bolts, nuts and screws shall be drawn tight. Electrical wiring shall be provided with adequate slack to relieve strain, and the wiring shall not be abraded, poorly joined, or loose at terminals. Rivets shall be tightly driven and fully peened.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth herein, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of tests. Tests shall be classified as first article (see 4.4) and quality conformance (see 4.5).

4.3 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at the prevailing ambient temperature and humidity conditions.

4.4 First article test. One tower, representative of production equipment, and various components as needed, shall be subjected to all the tests specified herein. The samples shall be identified with the manufacturer's part number and such other information as is required by this specification.

4.4.1 Pre-erection inspection tests. The tower shall be given a thorough mechanical and visual inspection and test to determine that the quality of all materials and workmanship are in compliance with all the requirements of this specification. Particular attention should be given to the following:

a. Choice and quality of materials.

b. Nameplates, identification markings and labels.

c. Finishes.

d. Welded joints.

e. Check of mounting means.

f. Check of corrosion prevention.

g. Check of safety features.

h. Loose fastening and securing devices or parts.

i. Grounding connections.

j. Over-all dimensions check.

k. Other visual defects.

4.4.2 Assembly. The contractor shall assemble one complete tower to insure a proper fit of all members.

4.4.2.1 Post-erection inspection tests. An erected tower shall be subjected to a careful mechanical inspection to insure that the requirements of this specification have been met. Particular reference shall be given to choice and quality of materials. Such inspection shall demonstrate the accuracy of fit and the correctness of dimensional data. All sections shall fit easily without having to be forced and without bending or straining of any part. Particular attention shall be given to the following points:

a. The fit of members and parts in their respective positions.

b. Accessibility of parts for servicing and inspection.

c. Check of mounting means.

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4.5 Quality conformance tests. The quality conformance tests shall consist of those tests listed in 4.4.1.

5. PREPARATION FOR DELIVERY

5.1 Disassembly. Prior to the preservation, packaging and packing operation, the tower shall be disassembled to the extent necessary to permit the required preservation and packaging and to facilitate packing in the most compact and economical manner.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.2.1 Level A. The disassembled tower components shall be preserved and packaged in accordance with Method III of MIL-P-116. Components shall be segregated as to like items. Where identification by visual observation would be difficult, segregated items shall be identified. Where necessary for physical protection during packing, the segregated items shall be cushioned, wrapped, bundled, or boxed as necessary to afford such protection.

5.2.1.1 Hardware. Hardware (nuts, bolts, washers, brackets, etc.) and structural support members shall be segregated and identified as required and intermediate packaged in boxes conforming to PPP-B-621, class 2, style 4, type 3 load, or PPP-B-601, overseas type, style optional, type 3 load.

5.2.1.2 Technical publications. Technical publications to be packed with towers shall be preserved by Method IC-1, MIL-P-116.

5.2.2 Level C. The complete tower shall be preserved in accordance with the supplier's standard practice.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. The main tower sections, structural support components and technical publications for each complete tower, preserved and packaged in compliance with 5.2.1, above, shall be packed in crates conforming to MIL-C-104, type I, class 2, style A. Packing for the complete tower will be accomplished in as few crates as practical. The packaged components shall be positioned so that the center of balance will be as near the center of each crate as possible. Cushioning, blocking, bracing and anchoring to prevent movement and subsequent damage to the items and crate shall be in accordance with MIL-STD-1186. Steel strapping shall conform to QQ-S-781, type I, heavy duty, 1 1/4 inch wide X .044 inch thick, finish B.

5.3.2 Level C. The complete tower shall be packed in a manner to insure arrival at destination in satisfactory condition and which will be acceptable to carriers at the lowest rates. Packing shall comply with ICC Uniform Freight Classification Rules or National Motor Freight Classification Rules, according to the mode of transportation.

5.4 Marking. Interior packages, bundles and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tower covered by this specification is intended to be

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used as an extension tower to increase the height of the following basic radar towers.

a. Antenna Tower AB-199A( ) FPS-3.

b. Tower AB-259A/FPS-6.

c. Tower AB-563/GPS.

6.2 Ordering data.

a. Title, number and date of this specification.

b. Level of preservation and packaging, and level of packing required (see 5.2 and 5.3).

c. Color of finish paint (see 3.7.2).

6.3 Asterisks. Due to the extensiveness of the revisions asterisks were not used to denote changes from the previous issue.

6.4 Drawing changes. In the event it becomes necessary to change the drawings (see 3.2) in order to achieve the performance requirements of this specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes shall be processed in accordance with the provisions of the changes clause of the contract.

Custodian:
Air Force - 84

Preparing Activity:
Air Force - 84

Project Number:
5445-F015

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-9640C (USAF) Tower AB-305( ) FPS

NAME OF ORGANIZATION AND ADDRESS
Warner Robins ALC/MMEME
Robins AFB, Ga. 31098

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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